

The diagram illustrates a 1D chain with two domains, D_1 (light blue) and D_2 (light orange), separated by a vertical dashed line at site index 11. The chain consists of sites indexed by j . Sites are represented by blue circles (v) and red squares (w). Vertical lines of different colors (red, orange, blue) represent excitations. The left domain D_1 contains excitations labeled $|\mathcal{L}\rangle$, $|S_1\rangle$, and $|\mathcal{R}\rangle$. The right domain D_2 contains excitations labeled $|\mathcal{L}\rangle$, $|S_1\rangle$, and $|\mathcal{R}\rangle$.

Energy E

Figure 3 is a line plot showing the squared norm of the wave function, $|\psi_j|^2$, versus Site j for three different states: $|\mathcal{L}\rangle$ (red), $|\mathcal{S}_1\rangle$ (orange), and $|\mathcal{R}\rangle$ (blue). The x-axis ranges from 0 to 22, and the y-axis ranges from 0.0 to 0.6. The plot shows that the wave functions are localized at different sites: $|\mathcal{L}\rangle$ is localized at $j=0$, $|\mathcal{S}_1\rangle$ is localized at $j=11$, and $|\mathcal{R}\rangle$ is localized at $j=22$.